

Bonded Authority Tokens

Co-Signed Paired Tokens with an Atomic, Append-Only State Register for Live Revocation of Agent Authority

Field	Value
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Status	Technical disclosure draft for attorney review — not a filed application
Reference impl.	axiom_event_token/bonded_pair.py

Two tokens are minted together, each cryptographically referencing the other. Their authority does not live in the token bytes — it lives in an append-only, hash-chained state register. Presenting either half to a gate resolves the pair's **current** state, so authority is checked against live state, not an inherited grant. Flipping the pair to REVOKED ends the primary's authority instantly — **without rotating the primary or its signing key**.

1 • Abstract

Systems and methods are disclosed for governing the authority of software agents using a bonded pair of cryptographic tokens. Two tokens — a primary and a mirror — are minted together, share a common pair identifier, and each carries a deterministic reference to the other. Token authority state is stored not in the signed token bytes but in an append-only, hash-chained ledger maintained by a single writer. A gate presented with either half consults the ledger for the pair's current state and grants, modifies, or refuses the requested action accordingly. A state transition to a terminal REVOKED state ends the primary's authority atomically and immediately, without rotating the primary token or its signing key. The primitive supports action-with-live-monitor, two-party atomic commit, and long-lived authorization with instant revocation.

2 • Field

Authorization and access control for autonomous and agentic AI systems; cryptographic tokens; revocation; tamper-evident audit ledgers; runtime governance of agent actions.

3 • Background & Problem

Conventional bearer credentials (API keys, OAuth tokens, JWTs) bind authority into the token itself. Revoking such a credential generally requires rotating a key or maintaining an out-of-band denylist, and a long-lived grant cannot be atomically paired with a live monitor. For agentic systems — where authority must be narrow, observable, and revocable mid-task — this is inadequate: a compromised or drifting agent keeps acting until its key is rotated, and there is no second party with standing to halt it without tearing down the grant.

4 • Summary of the Invention

The bonded authority token separates **identity** (the signed token bytes) from **authority state** (an external, append-only register):

Element	Description
Bonded pair	Two tokens minted together via <code>mint_pair()</code> : a primary (AXIOM-BP-<id>-A) and a mirror (-B), sharing a <code>pair_id</code> , each referencing the other's <code>token_id</code> .
Independent signing	Each token is HMAC-SHA256 signed under <code>axiom-bonded-pair-token-v1</code> ; the ledger under <code>axiom-bonded-pair-ledger-v1</code> .
State register	Authority state (ACTIVE_VALIDATED, ACTIVE_PENDING, SUSPENDED, REVOKED, EXPIRED) lives in a hash-chained ledger, not in the tokens. <code>transition()</code> is the only mutation surface.
Live resolution	A gate presented either half resolves the pair's current state from the ledger before permitting an action.
Revocation w/o rotation	Holding the mirror is sufficient to flip the pair to REVOKED — the primary and its key are never rotated.
Tamper-evidence	The ledger is hash-chained; a single byte flip breaks <code>verify_chain()</code> from that entry forward.

5 • Detailed Embodiments

5.1 Action + live monitor. The primary carries the command to execute; the mirror is held by a live security monitor. If the monitor observes unsafe behavior it flips the pair to REVOKED, short-circuiting the primary's authority mid-action — no key rotation, no redeploy.

5.2 Two-party atomic commit. Both halves must transition together; transition() is the single mutation surface, so neither party can advance authority unilaterally. Useful for dual-control / four-eyes operations.

5.3 Long-lived authorization with live revocation. The primary is a durable grant; the mirror is the revocation handle. Authority can be ended instantly by flipping the register, without rotating the long-lived grant or its signing key — the property conventional bearer tokens lack.

6 • Representative Data Flow

```
mint_pair(payload_primary, payload_mirror)
→ primary AXIOM-BP-<pid>-A (role=primary, partner=-B) [HMAC]
→ mirror AXIOM-BP-<pid>-B (role=mirror, partner=-A) [HMAC]
ledger.init_pair(pid) → state = ACTIVE_VALIDATED (chained entry)
gate(token) → ledger.current_state(pid) → ACTIVE_VALIDATED → ALLOW
ledger.transition(pid, REVOKED) [present mirror] → chained, signed entry
gate(primary) → ledger.current_state(pid) → REVOKED → DENY (key unchanged)
```

7 • Distinction From Conventional Systems

Conventional	Limitation	Bonded token
API key / bearer token	Revoke = rotate key / denylist	Revoke = flip register; key never rotates
JWT (state in token)	Stale until expiry; no live state	State is an external live register; resolved per request
OAuth grant	No co-bound monitor with standing to halt	Mirror halts the primary atomically
Audit log	Often mutable / after-the-fact	Append-only, hash-chained; verify_chain() detects any flip

8 • Representative Claim Seed Set

Drafting note: claim seeds for counsel; narrow/broaden after prior-art review.

Independent system claim. A system for governing agent authority, comprising: one or more processors; a token minting module configured to mint a bonded pair of tokens minted together and sharing a pair identifier, a first (primary) token and a second (mirror) token each carrying a deterministic reference to the other and an independent cryptographic signature; an append-only, hash-chained state register storing an authority state of the pair separately from the token bytes; and a gate configured to, upon presentation of either token, resolve the current authority state of the pair from the register and permit, modify, or refuse a requested action based thereon, wherein a transition of the register to a revoked state terminates the authority of the primary token without rotating the primary token or its signing key.

Dependent seeds. (2) wherein the mirror token is sufficient to effect the revoked transition. (3) wherein both tokens must transition together via a single mutation surface (two-party commit). (4) wherein the register is hash-chained such that altering any entry breaks verification from that entry forward. (5) wherein the mirror is held by a monitoring process that flips the pair upon detecting an unsafe action. (6) wherein

each token and each register entry is HMAC-signed under a distinct derived key. (7) wherein the authority state is one of active, pending, suspended, revoked, or expired. (8) wherein the agent is an autonomous AI agent and the action is a tool call.

Independent method claim. A method comprising: minting a primary and a mirror token together with a shared pair identifier and mutual references; recording an initial authority state in an append-only hash-chained register; receiving a request accompanied by either token; resolving the current state from the register; and permitting or refusing the request based on the state, wherein presenting the mirror transitions the register to a revoked state that terminates the primary's authority without rotating the primary or its key.

Computer-readable medium claim. A non-transitory medium storing instructions that cause the operations of the method claim.

9 · Implementation Notes

HMAC-SHA256 over canonical JSON; tokens and ledger signed under distinct namespaces (axiom-bonded-pair-token-v1, axiom-bonded-pair-ledger-v1); append-only JSONL ledger with a single writer; `verify_chain()` validates the hash chain. This is a co-signed token pair with an atomic state register — not entanglement; the design is defensible to an auditor on those terms.

This document should be reviewed by patent counsel before any public disclosure or filing.